

# BIOINFORMATICS TO AI AGENTS

A Complete Learning Roadmap — RNA-seq to Multiomics to AI Products

Bulk RNA-seq → Variant Calling (GATK) → ATAC-seq → ChIP-seq

scRNA-seq → Spatial Transcriptomics → Multiomics → AI Agents

10 Structured Phases | ~18–24 Months | Career & Product Ready

Compiled from: Nature Methods, Bioinformatics, PubMed, arXiv, GATK Docs, Bioconductor — 2025

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## **Phase 0 — Foundations — Linux, Python/R, Git, Statistics**

Duration: ~4 weeks

## **Phase 1 — Bulk RNA-seq Analysis — QC to Differential Expression**

Duration: ~6 weeks

## **Phase 2 — Variant Analysis — WGS/WES, GATK, Annotation**

Duration: ~6 weeks

## **Phase 3 — ChIP-seq & Epigenomics**

Duration: ~4 weeks

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Duration: ~4 weeks

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Duration: ~8 weeks

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Duration: ~4 weeks

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Duration: ~6 weeks

## **Phase 8 — AI/ML for Bioinformatics**

Duration: ~6 weeks

## **Phase 9 — AI Agents for Omics — Products & Career**

Duration: ~8 weeks

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*How to use this document: Each phase builds on the previous. Study the listed resources, run the listed tools on public datasets, then complete the mini-project before moving on. All datasets referenced are freely available from GEO, SRA, ENCODE, or 10x Genomics.*

## Why this comes first

Every downstream bioinformatics tool runs on the Linux command line. Python and R are used to write analysis scripts, handle data, and produce figures. Without these foundations you will struggle to even install tools, let alone interpret results.

## What to learn

- Linux / Bash: navigation, file permissions, pipes, grep/awk/sed, loops, screen/tmux
- Python: pandas, numpy, matplotlib, seaborn, Biopython, subprocess
- R: tidyverse, ggplot2, Bioconductor basics, DESeq2 data structures
- Git & GitHub: version control for your scripts and pipelines
- Statistics: distributions, t-test, FDR/p-value correction, PCA intuition, dimensionality reduction
- HPC/cluster basics: SLURM or PBS job submission, conda environments

## Best free resources

- › The Unix Shell — Software Carpentry ([swcarpentry.org/shell-novice](https://swcarpentry.org/shell-novice))
- › Python for Biologists — [pythonforbiologists.org](https://pythonforbiologists.org)
- › R for Data Science — [r4ds.had.co.nz](https://r4ds.had.co.nz)
- › Statistics for Genomics — StatQuest with Josh Starmer (YouTube — highly recommended)
- › HBC Training Harvard — [hbctraining.github.io/main](https://hbctraining.github.io/main)

## Mini-project

Write a Bash script that downloads a FASTQ file from SRA using fasterq-dump, runs FastQC on it, and saves the report. Push the script to GitHub.

## What bulk RNA-seq measures

Bulk RNA-seq measures the average gene expression across millions of cells in a sample. It is the most widely used transcriptomics method in academia and industry, and is the logical starting point because its analysis pipeline is linear and well-documented.

## The analysis pipeline — step by step

### Step 1: Quality Control

Run FastQC on raw FASTQ files. Look at per-base quality, GC content, adapter content. If quality is poor, use Trimmomatic or Cutadapt to trim adapters and low-quality bases.

### Step 2: Alignment

Map reads to the reference genome using STAR (preferred for RNA-seq) or HISAT2. This produces BAM files. Check alignment rates — >75% is expected for good data.

### Step 3: Quantification

Count how many reads map to each gene using featureCounts (Subread) or HTSeq-count. Alternatively use pseudo-alignment (Salmon or Kallisto) — faster and increasingly preferred.

### Step 4: Quality Assessment

Assess sample clustering with PCA. Check for outlier samples. Look at library size distributions and rRNA contamination.

### Step 5: Differential Expression

Use DESeq2 (R) or edgeR to identify genes that are significantly different between conditions. Understand the concept of normalisation (TMM, DESeq2 size factors), dispersion estimation, and negative binomial modelling.

### Step 6: Downstream Analysis

Gene Ontology (GO) enrichment, KEGG pathway analysis using clusterProfiler. Heatmaps of top DE genes. Volcano plots. GSEA (gene set enrichment analysis).

**Tools:** FastQC | Trimmomatic | STAR | HISAT2 | Salmon | Kallisto | featureCounts | DESeq2 | edgeR | clusterProfiler | MultiQC

## Best resources

- › STAR aligner manual — [github.com/alexdobin/STAR](https://github.com/alexdobin/STAR)
- › DESeq2 vignette — [bioconductor.org/packages/DESeq2](https://bioconductor.org/packages/DESeq2)
- › RNA-seq workflow — Bioconductor F1000 paper (Love et al. 2015)
- › HBC Training RNA-seq course — [hbctraining.github.io/DGE\\_workshop](https://hbctraining.github.io/DGE_workshop)
- › StatQuest DESeq2 videos — YouTube (Josh Starmer)
- › ENCODE RNA-seq standards — [encodeproject.org/data-standards/rna-seq](https://encodeproject.org/data-standards/rna-seq)

## Public dataset to practice on

GEO accession GSE96870 (mouse brain RNA-seq, clean 2-condition design). Download with SRA-tools, run the full pipeline, reproduce the DE gene list.

## Mini-project

Full pipeline on GSE96870: QC -> align with STAR -> quantify with featureCounts -> DESeq2 differential expression -> GO enrichment -> volcano plot. Document everything in an R Markdown or Jupyter notebook.

## What variant analysis finds

Variant calling identifies positions in the genome where an individual's DNA differs from the reference — SNPs (single nucleotide polymorphisms), indels (insertions/deletions), copy number variants (CNVs), and structural variants (SVs). GATK (Genome Analysis Toolkit) from the Broad Institute is the gold-standard pipeline for germline and somatic variant calling.

## The GATK Best Practices pipeline

### 1. Pre-processing

FASTQ -> BWA-MEM2 alignment -> SAM to BAM (samtools) -> Mark duplicates (GATK MarkDuplicates or Picard) -> Base Quality Score Recalibration (BQSR). BQSR corrects systematic errors in base quality scores using known variant databases (dbSNP).

### 2. Germline variant calling

GATK HaplotypeCaller in GVCF mode per sample -> GenomicsDBImport (for cohorts) -> GenotypeGVCFs -> Variant Quality Score Recalibration (VQSR) or hard filtering for small datasets. Output: VCF file.

### 3. Somatic variant calling

For tumour-normal pairs use Mutect2. Filter with FilterMutectCalls. Orientation bias filtering for FFPE samples.

### 4. Variant annotation

Annotate VCF with functional impact using ANNOVAR, SnpEff, or VEP (Ensembl). Add population frequencies from gnomAD, ClinVar pathogenicity, and SIFT/PolyPhen scores.

### 5. Variant filtering & interpretation

Apply ACMG classification guidelines for clinical variants. Use IGV (Integrative Genomics Viewer) to visually validate candidate variants.

### 6. Copy number & structural variants

GATK CNV pipeline (denoising coverage) for CNVs. DELLY, Manta, or Lumpy for structural variants from WGS data.

**Tools:** BWA-MEM2 | GATK4 | Picard | samtools | bcftools | ANNOVAR | SnpEff | VEP | IGV | DELLY | Manta | DeepVariant

## Best resources

- › GATK Best Practices — [gatk.broadinstitute.org/hc/en-us/sections/360007226651](https://gatk.broadinstitute.org/hc/en-us/sections/360007226651)
- › GATK documentation — [broadinstitute.github.io/picard](https://broadinstitute.github.io/picard)
- › GATK4 workshop tutorials — [github.com/broadinstitute/gatk](https://github.com/broadinstitute/gatk)
- › ENCODE variant calling standards — [encodeproject.org](https://encodeproject.org)
- › Bioinformatics clinician guide — PMC9870988 (J Thorac Oncol 2022)
- › DeepVariant paper — Nature Biotechnology 2018 (Poplin et al.)

## Mini-project

Use the GATK bundle's NA12878 test data (available at [gs://genomics-public-data/](https://gs://genomics-public-data/)). Run the germline pipeline from FASTQ to annotated VCF. Compare your results to the Genome in a Bottle truth set. Calculate precision/recall.

## What ChIP-seq measures

ChIP-seq (Chromatin Immunoprecipitation followed by sequencing) identifies where proteins (transcription factors, histone modifications) bind to DNA across the genome. It reveals regulatory elements — promoters, enhancers, silencers — and epigenetic marks like H3K27ac (active enhancers) and H3K4me3 (active promoters).

## The ChIP-seq pipeline

### 1. QC and alignment

FastQC on raw reads -> Trim with Trimmomatic -> Align with Bowtie2 -> Filter low-quality and duplicate reads. Check library complexity with NRF (Non-Redundant Fraction).

### 2. Peak calling

MACS2 or MACS3 calls peaks — genomic regions enriched for ChIP signal. For transcription factors: narrow peaks. For histone marks: broad peaks. Always run with matched input control to distinguish true signal from background.

### 3. Quality metrics

Calculate FRiP score (Fraction of Reads in Peaks) — >1% TF, >10% histone. Use phantompeakqualtools to compute NSC and RSC strand cross-correlation metrics.

### 4. Downstream analysis

Annotate peaks to nearest gene with ChIPseeker or HOMER. Motif analysis using HOMER findMotifsGenome or MEME-ChIP. Differential binding analysis with DiffBind. Visualise with deepTools bamCompare, plotHeatmap, plotProfile.

**Tools:** Bowtie2 | MACS2/3 | HOMER | deepTools | ChIPseeker | DiffBind | phantompeakqualtools | MEME-ChIP | IGV

## Best resources

- › ENCODE ChIP-seq pipeline v2 — [github.com/ENCODE-DCC/chip-seq-pipeline2](https://github.com/ENCODE-DCC/chip-seq-pipeline2)
- › deepTools documentation — [deeptools.readthedocs.io](https://deeptools.readthedocs.io)
- › MACS2 paper — Genome Biology 2012 (Zhang et al.)
- › HBC ChIP-seq training — [hbctraining.github.io/Intro-to-ChIPseq](https://hbctraining.github.io/Intro-to-ChIPseq)
- › ChIPseeker vignette — [bioconductor.org/packages/ChIPseeker](https://bioconductor.org/packages/ChIPseeker)

## Mini-project

Download ENCODE H3K27ac ChIP-seq data for a cell line (e.g. K562, available at [encodeproject.org](https://encodeproject.org)). Run MACS2, annotate peaks, identify active enhancers and their target genes.

## What ATAC-seq measures

ATAC-seq (Assay for Transposase-Accessible Chromatin using sequencing) maps open chromatin regions genome-wide — places where the DNA is accessible for transcription factor binding. It is faster and requires far fewer cells than ChIP-seq. Critically, it bridges RNA-seq (what genes are expressed) and ChIP-seq (where regulatory proteins bind), making it essential for understanding gene regulation.

## The ATAC-seq pipeline

### 1. QC and pre-processing

Paired-end reads are required. Run FastQC. Trim Nextera adapters with Trimmomatic (ILLUMINACLIP:NexteraPE-PE.fa:2:30:10). Check insert size distribution — should show nucleosome-free (~150 bp) and mono-nucleosome (~200 bp) peaks.

### 2. Alignment

Align to genome with Bowtie2 (--very-sensitive, -X 2000). Remove mitochondrial reads (high background), duplicates, and reads with low MAPQ. Shift reads by +4 bp (Tn5 insertion correction) using ATACseqQC or deepTools.

### 3. Peak calling

MACS2 with --nomodel --shift -100 --extsize 200 for ATAC. Alternatively use Genrich (ATAC-specific caller). Generate a consensus peak set across samples.

### 4. Quality metrics

FRIP score, TSS enrichment score (key metric — >6 is good), nucleosomal signal ratio. Use ATACseqQC R package for comprehensive QC.

### 5. Downstream analysis

Differential accessibility with DiffBind or DESeq2 on count matrices. TF footprinting (TOBIAS, HINT-ATAC) identifies bound TFs in open chromatin. Link open chromatin to gene expression using RNA-seq co-analysis.

**Tools:** Bowtie2 | Trimmomatic | MACS2 | Genrich | ATACseqQC | deepTools | TOBIAS | HINT-ATAC | DiffBind | ChromHMM

## Best resources

- › ENCODE ATAC-seq pipeline — [github.com/ENCODE-DCC/atac-seq-pipeline](https://github.com/ENCODE-DCC/atac-seq-pipeline)
- › ATAC-seq best practices — [nature.com/articles/s41596-022-00692-9](https://www.nature.com/articles/s41596-022-00692-9) (Nat Protocols 2022)
- › ATACseqQC — [bioconductor.org/packages/ATACseqQC](https://bioconductor.org/packages/ATACseqQC)
- › TOBIAS footprinting — [github.com/loosolab/TOBIAS](https://github.com/loosolab/TOBIAS)
- › Buenrostro et al. 2013 — original ATAC-seq paper (Nature Methods)

## Mini-project

Download ENCODE ATAC-seq data for PBMC or a cancer cell line. Run the full pipeline, identify open chromatin at promoters vs enhancers, and do TF motif enrichment at ATAC peaks.

## What scRNA-seq adds over bulk RNA-seq

Bulk RNA-seq gives the average expression across all cells in a sample. scRNA-seq resolves this into individual cell profiles — revealing rare cell types, cell state transitions, lineage trajectories, and cell-to-cell heterogeneity that is invisible in bulk data. This phase is the longest because it has the most analysis steps and the most active tooling ecosystem.

## The scRNA-seq pipeline

### 1. Raw data processing

If using 10x Genomics: run Cell Ranger count to align reads and generate a cell-by-gene count matrix (barcodes.tsv, features.tsv, matrix.mtx). For other platforms (Parse, BD Rhapsody) use their respective demultiplexing pipelines.

### 2. Quality control

Load into Scanpy (Python) or Seurat (R). Filter cells: remove cells with too few genes (empty droplets), too many genes (doublets), or high mitochondrial read fraction (dying cells). Use Scrublet or DoubletFinder to computationally detect doublets.

### 3. Normalisation & feature selection

Normalise library size (scraper, or simple CPM). Log-transform. Select highly variable genes (HVGs) — typically top 2000-5000 genes that drive biological variation.

### 4. Dimensionality reduction

PCA on HVGs. Compute UMAP or t-SNE for visualisation (UMAP is preferred — preserves global structure better). Understand that UMAP is for visualisation only — don't over-interpret distances.

### 5. Clustering

Graph-based clustering (Leiden or Louvain algorithm on a k-nearest neighbour graph). The resolution parameter controls the number of clusters — test multiple values.

### 6. Cell type annotation

Find marker genes for each cluster with Wilcoxon rank-sum test. Match markers to known cell type signatures using CellMarker database, PanglaoDB, or CellTypist. Use reference-based annotation (SingleR, scANVI) for automated labelling.

### 7. Differential expression

Pseudobulk DE is the gold standard — aggregate cells per sample per cell type, then use DESeq2. Avoid simple Wilcoxon between clusters for multi-sample studies.

### 8. Trajectory analysis

Infer differentiation trajectories with Monocle3, PAGA, or scVelo (RNA velocity using spliced/unspliced read ratios). Use Slingshot for lineage reconstruction.

### 9. Cell communication

Infer ligand-receptor interactions between cell types using CellChat, CellPhoneDB, or LIANA. Understand which cells are 'talking' to each other.

**Tools:** Cell Ranger | Scanpy | Seurat | scraper | Scrublet | DoubletFinder | CellTypist | SingleR | Monocle3 | scVelo | PAGA | CellChat | CellPhoneDB | Harmony | scVI | LIANA

## Best resources

- › Scanpy tutorials — [scanpy.readthedocs.io/en/stable/tutorials.html](https://scanpy.readthedocs.io/en/stable/tutorials.html)
- › Seurat vignettes — [satijalab.org/seurat/articles/pbmc3k\\_tutorial](https://satijalab.org/seurat/articles/pbmc3k_tutorial)
- › Single-cell best practices book — [sc-best-practices.org](https://sc-best-practices.org) (free online textbook)
- › Bioconductor OSCA book — [bioconductor.org/books/release/OSCA](https://bioconductor.org/books/release/OSCA) (Orchestrating Single-Cell Analysis)
- › scVelo — [scvelo.readthedocs.io](https://scvelo.readthedocs.io)
- › CellChat — [github.com/jinworks/CellChat](https://github.com/jinworks/CellChat)
- › StatQuest scRNA-seq playlist — YouTube
- › 10x Genomics datasets — [10xgenomics.com/datasets](https://10xgenomics.com/datasets) (PBMC 3k, 10k — free download)

## Mini-project

Use the 10x PBMC 3k or 10k dataset. Run the full Scanpy pipeline: QC -> normalisation -> HVG selection -> PCA -> UMAP -> Leiden clustering -> marker gene identification -> cell type annotation -> pseudobulk DE. Then add RNA velocity with scVelo. Produce a publication-quality UMAP figure.

## What spatial transcriptomics adds

scRNA-seq loses the physical location of cells in tissue. Spatial transcriptomics preserves that spatial context — revealing how cell types are organised in tissue, identifying spatially variable genes, and mapping cell-cell communication in situ. Major platforms: 10x Visium (spot-level), Xenium/MERFISH (single-cell resolution), Slide-seq, and NanoString GeoMx.

## The spatial pipeline

### 1. Data processing

10x Visium: Space Ranger to align reads and map to tissue spots. Output: count matrix with spatial coordinates (x,y per spot).

### 2. QC and normalisation

Filter spots with too few counts. Normalise with scran or SCTransform. Identify spatially variable genes with SpatialDE or SPARK.

### 3. Spatial clustering

Cluster spots by gene expression (same methods as scRNA-seq). Visualise clusters overlaid on H&E; tissue image.

### 4. Cell type deconvolution

Each Visium spot contains ~5-20 cells — use RCTD, SPOTlight, or cell2location to deconvolve spot composition using a scRNA-seq reference atlas.

### 5. Spatial domain identification

BayesSpace or STAGATE identify coherent tissue domains. These correspond to known anatomical regions (tumour core, stroma, immune infiltrate).

### 6. Integration with scRNA-seq

Map scRNA-seq cell types onto spatial coordinates using Tangram or Seurat integration. Build a complete spatial cell atlas.

**Tools:** Space Ranger | Squidpy | Seurat (spatial) | SpatialDE | SPARK | RCTD | cell2location | BayesSpace | STAGATE | Tangram

## Best resources

- › Squidpy tutorials — [squidpy.readthedocs.io](https://squidpy.readthedocs.io)
- › Seurat spatial vignette — [satijalab.org/seurat/articles/spatial\\_vignette](https://satijalab.org/seurat/articles/spatial_vignette)
- › cell2location — [cell2location.readthedocs.io](https://cell2location.readthedocs.io)
- › 10x Visium public datasets — [10xgenomics.com/datasets](https://10xgenomics.com/datasets) (mouse brain, human breast cancer)
- › Spatial omics review — Nat Rev Genet 2022 (Moses & Pachter)

## Mini-project

Download the 10x Visium human breast cancer dataset. Run Space Ranger, then Squidpy for spatial clustering. Deconvolve spots using a public breast cancer scRNA-seq atlas. Map immune cell infiltration patterns onto the tissue image.

## What multiomics integration achieves

No single assay gives the full picture. Multiomics combines RNA-seq, ATAC-seq, ChIP-seq, proteomics, and metabolomics to understand how the genome is regulated, which genes are expressed, what proteins are made, and what metabolites result. This is the frontier of modern precision medicine and drug discovery.

## Key multiomics frameworks

### scRNA + scATAC (10x Multiome)

Joint profiling of gene expression and chromatin accessibility in the same cell. Use ArchR or Signac (R) for scATAC-seq analysis. Link ATAC peaks to gene expression (peak-gene links) to identify cis-regulatory elements.

### CITE-seq (RNA + Protein)

Simultaneously measure RNA and surface protein expression (ADT). Use weighted nearest neighbour (WNN) in Seurat to integrate both modalities for clustering. Identify cell types that are ambiguous by RNA alone.

### Bulk multiomics integration

Integrate bulk RNA-seq, ChIP-seq, and ATAC-seq to understand transcription factor regulatory networks. Use VIPER, DoRothEA, or SCENIC for TF activity inference.

### Multi-omics factor analysis (MOFA)

MOFA+ (Python/R) decomposes variation across multiple omics layers into latent factors. Identifies coordinated regulatory programmes. Apply to datasets with RNA + ATAC + proteomics.

### Proteomics integration

Integrate mass spectrometry-based proteomics with RNA-seq. Use Perseus for proteomics analysis. Correlate protein and RNA abundance to identify post-transcriptional regulation.

### Metabolomics integration

Combine metabolomics with transcriptomics using COSMOS or MixOmics. Map metabolite changes to pathway activity.

**Tools:** ArchR | Signac | Seurat WNN | MOFA+ | SCENIC | DoRothEA | VIPER | MixOmics | MultiVI | StabMap | Perseus

## Best resources

- › ArchR — [archrproject.com](https://archrproject.com)
- › Signac — [stuartlab.org/signac](https://stuartlab.org/signac)
- › MOFA+ — [biofam.github.io/MOFA2](https://biofam.github.io/MOFA2)
- › SCENIC — [scenic.aertslab.org](https://scenic.aertslab.org)
- › Multi-omics integration review — Front Line Genomics Multiomics Playbook 2025
- › Nature Methods multiomics benchmarking papers (2024-2025)

## Mini-project

Download a 10x Multiome dataset (RNA + ATAC in same cells) from 10x Genomics website (e.g. PBMC multiome). Run ArchR for ATAC analysis, Seurat for RNA, then link peaks to genes. Identify TF motifs at accessible peaks and connect them to gene expression.

## Why AI is transforming bioinformatics

Classical statistics handles well-defined questions (is gene X differentially expressed?). Machine learning handles pattern recognition in high-dimensional data — predicting drug response from gene expression, classifying tumour subtypes from multiomics, or imputing missing measurements. Foundation models (scGPT, Geneformer) are now pre-trained on millions of cells and can be fine-tuned for specific tasks.

## Key AI/ML areas for bioinformatics

### Classical ML on omics data

Scikit-learn: logistic regression, random forest, SVM, XGBoost for classification tasks (tumour vs normal, responder vs non-responder). Feature selection: LASSO, elastic net for high-dimensional gene expression data. Survival analysis: Cox PH regression, survival random forests.

### Deep learning for genomics

CNNs for sequence-to-function prediction (ENCODE regulatory elements). RNNs/LSTMs for sequence modelling. Transformers for chromatin state prediction (Enformer, Basenji2).

### Foundation models for single-cell

scGPT — pre-trained on 33M cells, fine-tunable for cell annotation, perturbation prediction, batch correction. Geneformer — tokenises cells as gene rank sequences, predicts dosage-sensitive genes. Universal Cell Embeddings (UCE) — zero-shot cell type transfer.

### Variational autoencoders (VAE)

scVI and scANVI use deep VAEs for probabilistic modelling of scRNA-seq data — batch correction, data integration, semi-supervised annotation, imputation.

### Graph neural networks

Cell-cell communication modelling, protein-protein interaction networks, gene regulatory network inference using GNNs (DeePathology, CELL-E).

### Perturbation prediction

GEARS, scGPT, and CPA (Compositional Perturbation Autoencoder) predict gene expression responses to drug treatments or genetic perturbations — critical for drug discovery applications.

**Tools:** scikit-learn | PyTorch | scVI | scANVI | scGPT | Geneformer | Enformer | GEARS | CPA | XGBoost | PyG (PyTorch Geometric)

## Best resources

- › scGPT — [github.com/bowang-lab/scGPT](https://github.com/bowang-lab/scGPT)
- › scvi-tools — [scvi-tools.org](https://scvi-tools.org)
- › Geneformer — [huggingface.co/ctheodoris/Geneformer](https://huggingface.co/ctheodoris/Geneformer)
- › Deep learning genomics review — Nature Reviews Genetics 2019 (Zou et al.)
- › Enformer paper — Nature Methods 2021 (Avsec et al.)
- › fast.ai practical deep learning course — [fast.ai](https://fast.ai) (accessible, Python-based)

## Mini-project

Train a cell type classifier on the 10x PBMC dataset using scikit-learn (random forest). Then fine-tune scGPT for the same classification task. Compare accuracy. Use scVI for batch correction on a multi-sample dataset.

## What AI agents add

AI agents combine LLMs with domain tools to automate analysis pipelines that previously required expert bioinformaticians. They can receive a natural-language task, load a dataset, execute the right analysis steps, and return biological insights — democratising access to complex omics analysis. This is the frontier that creates commercial products.

## Key AI agents in omics (2025)

### CellAgent (arXiv 2407.09811)

Multi-agent framework with Planner/Executor/Evaluator roles. Powered by GPT-4. Accepts natural-language tasks and runs full scRNA-seq pipelines autonomously.

### CellAtria (Nature npj AI 2026)

Document-to-analysis automation. Reads a paper, extracts metadata, downloads the dataset, and runs Scanpy-based analysis end-to-end through a chatbot interface.

### CellVoyager (Nature Methods 2026)

Autonomous hypothesis generation. Takes a paper and dataset, generates and tests new biological hypotheses not explored by the original authors.

### OmniCellAgent (PMC12324537 2025)

Multi-agent system for precision medicine. Integrates scRNA-seq analysis, literature search, and a biomarker knowledge graph. Designed for clinicians with no coding background.

### AutoBA

Autonomous multi-omic agent. Requires only data path, description, and objective. Generates and executes analysis plans automatically across RNA-seq, ATAC-seq, and more.

### scBench (arXiv 2602.09063 2026)

Benchmark framework to evaluate your agent on 394 real scRNA-seq tasks across 6 platforms. Best model (Claude Opus 4.6) reaches 52.8% accuracy — use this to measure and improve your agent.

## How to build your own omics AI agent

- Define the scope: one assay type (e.g. scRNA-seq QC) before going general
- Wrap tools as callable functions: Scanpy functions become agent tools
- Choose your LLM backbone: Claude, GPT-4o, or local models (Llama 3.1)
- Implement Planner/Executor/Evaluator pattern (see CellAgent paper)
- Add domain knowledge: inject expert biology context into the system prompt
- Evaluate on scBench canonical evals: [github.com/latchbio/scbench](https://github.com/latchbio/scbench)
- Benchmark and iterate: measure accuracy per task category, fix weak spots
- Package as a product: REST API, web interface, or Nextflow pipeline wrapper

## Commercial product ideas

- Clinical variant interpretation SaaS: automated ACMG classification from VCF with AI rationale
- scRNA-seq analysis platform for wet-lab scientists (no-code, natural language interface)
- Drug response prediction tool: multiomics + ML model fine-tuned on CCLE/TCGA
- Spatial tissue analysis service for pathology labs using AI-powered cell type mapping

- AI literature mining + dataset analysis agent for pharma target discovery
- Automated QC and report generation pipeline as a containerised product (Docker + Nextflow)

## Career tracks this roadmap prepares you for

- Bioinformatics Scientist — NGS pipelines, variant calling, RNA-seq (pharma/biotech)
- Computational Biologist — algorithm development, multiomics, method papers (academia/biotech)
- Single-Cell & Multiomics Analyst — scRNA-seq, spatial, multiome (high demand 2025-2026)
- AI-Based Drug Discovery Scientist — ML on omics for target ID and biomarker discovery
- Clinical Bioinformatics Analyst — CAP/CLIA pipelines, variant interpretation (hospital/diagnostics)
- Cloud Bioinformatics Engineer — Nextflow/Snakemake + AWS/GCP pipelines (SaaS companies)
- Bioinformatics Software Developer — building tools, APIs, platforms (tech-bio startups)

## Best resources for AI agents in omics

- › CellAgent — [arxiv.org/abs/2407.09811](https://arxiv.org/abs/2407.09811)
- › CellAtria — [nature.com/articles/s44387-025-00064-0](https://nature.com/articles/s44387-025-00064-0)
- › CellVoyager — [nature.com/articles/s41592-026-03029-6](https://nature.com/articles/s41592-026-03029-6)
- › scBench — [arxiv.org/abs/2602.09063](https://arxiv.org/abs/2602.09063) & [github.com/latchbio/scbench](https://github.com/latchbio/scbench)
- › OmniCellAgent — [pmc.ncbi.nlm.nih.gov/articles/PMC12324537](https://pmc.ncbi.nlm.nih.gov/articles/PMC12324537)
- › LangChain / LlamaIndex — for building LLM agent frameworks in Python
- › Anthropic Claude API — for building production-grade biology agents

# Master Timeline at a Glance

| Phase | Topic                                                  | Duration                  | Key Output / Milestone                                         |
|-------|--------------------------------------------------------|---------------------------|----------------------------------------------------------------|
| 0     | Foundations<br>Linux, Python, R, Stats                 | 4 weeks                   | GitHub repo with scripts<br>Conda environments set up          |
| 1     | Bulk RNA-seq<br>QC to DE analysis                      | 6 weeks                   | DESeq2 DE gene list + GO enrichment<br>R Markdown report       |
| 2     | Variant Calling<br>GATK Best Practices                 | 6 weeks                   | Annotated VCF from NA12878<br>Precision/recall vs truth set    |
| 3     | ChIP-seq<br>Epigenomics                                | 4 weeks                   | MACS2 peak set + motif analysis<br>H3K27ac enhancer map        |
| 4     | ATAC-seq<br>Open Chromatin                             | 4 weeks                   | ATAC peak set + TF footprinting<br>Differential accessibility  |
| 5     | scRNA-seq<br>Cell-by-cell analysis                     | 8 weeks                   | Annotated UMAP + cell types<br>Pseudobulk DE + trajectory      |
| 6     | Spatial Transcriptomics<br>Tissue context              | 4 weeks                   | Cell type deconvolution map<br>Spatially variable genes        |
| 7     | Multomics Integration<br>RNA+ATAC+Protein              | 6 weeks                   | Peak-gene links + TF regulons<br>MOFA factor analysis          |
| 8     | AI/ML for Bioinformatics<br>Models & Foundation Models | 6 weeks                   | Trained classifier + scVI integration<br>scGPT fine-tuning     |
| 9     | AI Agents for Omics<br>Products & Career               | 8 weeks                   | Working agent evaluated on scBench<br>Portfolio + product demo |
| TOTAL |                                                        | ~56 weeks<br>(~14 months) | Career-ready portfolio<br>AI product prototype                 |

## Important notes on pacing

- Do not skip phases. Each phase builds the vocabulary and intuition needed for the next.
- Run every analysis on real public data. Reading is not enough — hands-on is essential.
- Keep a GitHub portfolio. Employers want to see working code, not just a CV of tool names.
- scRNA-seq (Phase 5) is the most time-consuming — do not rush it. It underpins Phases 6-9.
- The AI agent phase (Phase 9) only makes sense after you can evaluate the agent's biological output.
- You can compress the timeline by studying full-time, or stretch it by studying part-time.
- Join the Bioconductor Slack, scverse forum, and GATK forum for community support.

References & data sources: [GATK Best Practices \(Broad Institute\)](#) | [scBench arXiv 2602.09063](#) | [scRNA-seq best practices](#) [sc-best-practices.org](#) | [ENCODE project](#) [encodeproject.org](#) | [10x Genomics public datasets](#) | [Bioconductor OSCA book](#) | [StatQuest YouTube](#) | [BioSpace career outlook 2025](#) | [Nature Methods 2024-2026](#) | [PMC open-access literature](#)